CARD 1 OF 8 HOW OFTEN **QUESTION**

How often does NFPA 25 want someone to look at the **inside** of a sprinkler system's piping — and what can change that interval?

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CARD 2 OF 8 WHAT IT LOOKS FOR **QUESTION**

When you open the pipe for the internal assessment, **what are you actually looking for?**

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CARD 3 OF 8 MIC **QUESTION**

You open the end of a cross main and find **tubercles and slime**. What does NFPA 25 require next?

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CARD 4 OF 8 NEXT STEP **QUESTION**

The assessment finds **some buildup**. Does that automatically mean an **obstruction investigation**?

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*FSM · Sprinkler Fact Cards — NFPA 25 (2020) Internal Pipe Assessment · sheet 1 FRONT — print double-sided, **flip on LONG edge** · cut on the dashed lines*

CARD 2 OF 8 WHAT IT LOOKS FOR **ANSWER**

Foreign material, organic and inorganic. Scale, rust nodules (tubercles), slime, sludge, sand, rocks, debris.

What the Annex says it really is: mostly a **corrosion check**. Finding blockage material is a by-product.

Scope: you don't have to prove every foot of pipe is clean. The aim is **reasonable assurance** that corrosion and obstruction problems get spotted.

NFPA 25 (2020) · Ch. 14 · Annex A

On the job: you pull a 1" branch-line end cap and find orange crust and black water. That's exactly what this check exists to find.

CARD 1 OF 8 HOW OFTEN **ANSWER**

At least every 5 years.

Different interval: an **approved risk analysis** can set its own frequency, shorter or longer.

Clock reset (Annex): if the system gets drained for a renovation or tenant fit-out and the pipe is opened, inspected and **recorded** while it's down, that counts. The next 5 years start from that date.

NFPA 25 (2020) · Ch. 14 · Annex A

Trap: "We had a 5-year done on the check valves." Different item. The internal pipe assessment is its own task.

CARD 4 OF 8 NEXT STEP **ANSWER**

No. It has to be enough to block pipe or sprinklers. Then a full obstruction investigation follows.

Annex: most systems have *some* material or corrosion in them. A little is normal and not a trigger on its own.

If the investigation finds enough to obstruct: a **complete flushing program** run by qualified people.

NFPA 25 (2020) · Ch. 14 · Annex A

Trap: treating the 5-year assessment and the obstruction investigation as the same thing. The assessment is the routine look. The investigation is the deep dig it can lead to.

CARD 3 OF 8 MIC **ANSWER**

Test them for MIC (microbiologically influenced corrosion).

Tubercles are rust nodules growing on the pipe wall. **Slime** is a bacterial film. Both are signs bacteria may be eating the pipe from the inside.

Why it matters: MIC is what drills **pinhole leaks** in pipe that looks fine from the outside. Pinhole leaks are also on the list of things that trigger an obstruction investigation.

NFPA 25 (2020) · Ch. 14

Trap: scraping it out and closing the pipe back up. Finding it means testing it.

CARD 5 OF 8 MULTIPLE SYSTEMS **QUESTION**

A warehouse has **6 wet pipe systems**. Do all 6 get the internal assessment every 5 years?

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CARD 6 OF 8 HOW IT'S DONE **QUESTION**

How do you assess the inside of the pipe? Name the usual method and one alternative.

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CARD 7 OF 8 WHERE TO LOOK **QUESTION**

You only get to open the system in a couple of places. **Where should you look?**

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CARD 8 OF 8 OTHER TRIGGERS **QUESTION**

Apart from the 5-year assessment, name **four things** that make NFPA 25 call for an **obstruction investigation**.

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CARD 6 OF 8 HOW IT'S DONE **ANSWER**

Usual method (Annex): at the far end of a main, open the flushing connection and pull the **end fitting**, a **piece of branch line**, or a **sprinkler**, and look inside.

Alternative: a **video camera** on a push cable, fed in through the alarm, dry or preaction valve to see the riser, feed main and some cross mains.

Pulled a sprinkler? Put in a **new** one that matches the old one. Don't reinstall the old head.

Exempt: nonmetallic pipe (CPVC) doesn't need this assessment.

NFPA 25 (2020) · Ch. 14 · Annex A

CARD 5 OF 8 MULTIPLE SYSTEMS **ANSWER**

No. Every other system. Do 3 this cycle and the other 3 at the next one, so each system still gets looked at.

The catch: if foreign material turns up in **any** system in the building, **all** of them get assessed.

Why it works (Annex): the half you assess is treated as **representative** of every system in the building.

NFPA 25 (2020) · Ch. 14 · Annex A

Trap: the alternating rule is written for **wet** systems. Don't apply it to the dry systems in the same building without checking.

CARD 8 OF 8 OTHER TRIGGERS **ANSWER**

Any of these (the list has 15):

- **Plugged sprinklers**, or a plugged inspector's test
- **Pinhole leaks**
- Junk in the water during a **drain test**, or found in a dry valve, check valve or fire pump
- A system back in service after being shut down **more than 1 year**
- **Raw water** pumped in through the FDC
- **Broken public mains** nearby
- A **dry system** whose water now takes **50% longer** to reach the inspector's test than it did at acceptance
- A dry valve that trips falsely **a lot more than normal**

NFPA 25 (2020) · Ch. 14

CARD 7 OF 8 WHERE TO LOOK **ANSWER**

Where corrosion runs fastest (Annex):

- **Wet systems:** the **high points**, where trapped air sits.
- **Dry and preaction:** the **low points**, where water gets trapped.
- **Bulk piping** that sees a lot of fresh water flowing through it, like the supply piping near a **fire pump**.

Dry or preaction branch line: pick the one **farthest from the water source** that **doesn't** have the inspector's test valve on it.

NFPA 25 (2020) · Annex A (advisory)

Why air and water matter: corrosion needs both, and it's worst where they meet. A wet system's air pocket and a dry system's trapped puddle are where that happens.

sheet 2 BACK — columns are intentionally mirrored for long-edge duplex · card numbers must match front to back